

Attention Distillation: A Unified Approach to Visual Characteristics Transfer

Yang Zhou Xu Gao Zichong Chen Hui Huang*
 Visual Computing Research Center, CSSE, Shenzhen University

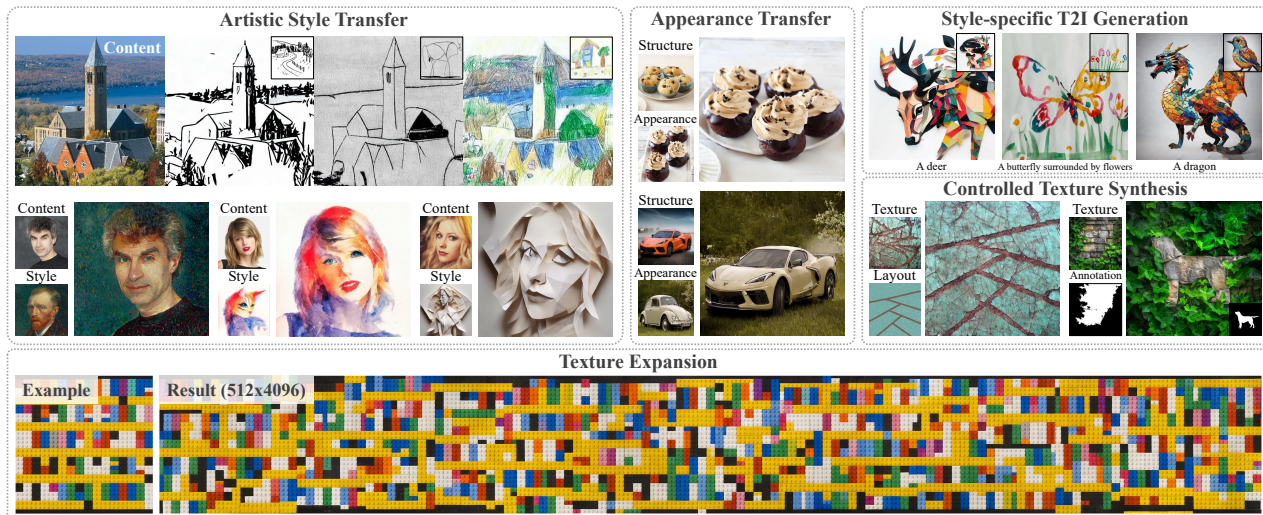


Figure 1. Given a reference image, our approach can faithfully reproduce its visual characteristics in synthesis, providing a unified framework for a wide range of example-based image synthesis applications, such as artistic style transfer, appearance transfer, style-specific text-to-image generation, and various texture synthesis tasks.

Abstract

Recent advances in generative diffusion models have shown a notable inherent understanding of image style and semantics. In this paper, we leverage the self-attention features from pretrained diffusion networks to transfer the visual characteristics from a reference to generated images. Unlike previous work that uses these features as plug-and-play attributes, we propose a novel attention distillation loss calculated between the ideal and current stylization results, based on which we optimize the synthesized image via backpropagation in latent space. Next, we propose an improved Classifier Guidance that integrates attention distillation loss into the denoising sampling process, further accelerating the synthesis and enabling a broad range of image generation applications. Extensive experiments have demonstrated the extraordinary performance of our approach in transferring the examples' style, appearance, and texture to new images in synthesis. Code is available at <https://github.com/xugao97/AttentionDistillation>.

1. Introduction

Synthesizing new images with visual elements, such as the style or texture, of an example image, is a long-standing yet challenging problem in computer graphics and vision. The key challenge lies in properly representing images' texture or style features. Traditional methods [6, 11, 15, 34, 37, 38, 42, 67] usually define textures as repeated local patterns and synthesize new textures by copying local patches from the source image. When it comes to style, an extensive yet more abstract visual characteristic than texture, new representations are required.

Thanks to the deep learning revolution, neural representations of visual features have emerged. One group of approaches performs texture or style-specific synthesis by matching the global distribution of deep features between the reference and the output. For example, the seminal work Gram loss [21, 22] regards the feature maps' statistics as the texture/style representation. Some other work optimizes deep features by minimizing Wasserstein distance [24] or adversarial discrimination loss [51, 52, 68]. However, matching global distributions lacks local perception, usually leading to conspicuous detail artifacts. Another group retakes

*Corresponding author.

the local patch strategy, optimizing output’s deep features through nearest-neighbor matching [40, 43, 69]. As local patches are unaware of global structure, these methods always require additional structural guidance.

Beyond the above progress, recent breakthroughs in large-scale diffusion models have sparked new representations for visual features, reaching a good balance between global and local consistency in synthesis [2, 7, 10, 25, 29, 70]. A key consensus of these techniques is that the *keys* (K) and *values* (V) from the self-attention module of pre-trained diffusion models characterize the appearance of the exemplar. Re-aggregating these features according to target *queries* (Q) by self-attention mechanism during new image generation can reproduce the visual characteristics of the reference. Although impressive results were achieved due to the robust performance of diffusion models, these methods still often suffer issues like insufficient stylization. We hypothesize three reasons for their limitations:

1) *Domain gap*. When two images differ significantly, the similarity between the target Q (queries of the synthesized image) and the source KV (keys and values of the exemplar) will become low and unreliable, leading to erroneous aggregation results. Techniques like AdaIN [27] and attention scale can partially mitigate this issue [2, 25].

2) *Error accumulation*. While the iterative sampling process in diffusion models can ameliorate large discrepancies between the target Q and the source KV , errors may also accumulate. As demonstrated in [56], features from different layers of diffusion models focus on distinct information, such as semantics and geometry. Incorrect matches will propagate errors to subsequent layers along the Markov chain and degrade the final image quality.

3) *Architectural limit*. The self-attention mechanism is implemented within the residual branch of the denoising network. Injecting self-attention features from the source may have a bounded influence on the target latent code, potentially diminishing their efficacy in the synthesis.

In this work, we still follow the assumption that the self-attention features in the denoising networks capture an image’s visual appearance. To address the above-mentioned limitations, we introduce a novel *Attention Distillation* (AD) loss, based on which we directly update the synthesized image through backpropagation. Specifically, we consider the output obtained by computing the attention between the target Q and the source KV as the *ideal* stylized result, and the original attention output represents the current stylization. We define attention distillation loss as the L1 distance between these two outputs and optimize the synthesized image through backpropagation in latent space. We simultaneously calculate the differences across various layers to avoid error accumulation. Such an optimization process gradually reduces the disparity between the target Q and the source KV , enhancing the accuracy of similarity

calculations and thus improving the final stylization.

Whereas our approach differs greatly from previous works that use self-attention features as plug-and-play attributes, the new attention distillation loss can also be integrated into the sampling process of diffusion models, functioning as an *improved Classifier Guidance*. Combined with the normal Classifier-Free Guidance, it further enables text-based controlled generation, and is also compatible with other conditioning technologies such as ControlNet, leading to broad image synthesis applications; see, *e.g.*, Fig. 1. Extensive experiments and comparisons with state-of-the-art methods have demonstrated our advantages.

In summary, our main contributions are as follows:

- We analyze the limitations of previous plug-and-play attention features methods and propose a novel attention distillation loss for reproducing the visual characteristics of a reference image, achieving notably superior results.
- We develop attention distillation guided sampling, an improved Classifier Guidance that integrates attention distillation loss into the denoising process, which significantly accelerates synthesis speed and enables a wide range of visual characteristics transfer and synthesis applications.

2. Related work

Our primary contribution lies in proposing a novel loss function that effectively transfers the visual characteristics of one image to another. This loss function has broad applications across various image synthesis tasks, including texture synthesis, style transfer, appearance transfer, and customized image generation based on text-to-image models. We review related work in these domains.

Texture synthesis focuses on generating images that resemble the original texture without repetition and artifacts. Traditional methods rely on parametric texture models [23, 32, 53] or sampling pixels/patches [14, 15, 37, 38, 62] to create new images. These methods excel with simple textures but often struggle with complex or high-resolution textures. Deep learning-based methods, utilizing convolution neural networks (CNNs), extract multilevel features that capture textures at varying scales. Gatys *et al.* [21] introduced Gram matrix, a second-order statistic of feature map to represent stationary textures. Histogram loss [47] and Sliced Wasserstein loss [16, 24] offer improved modeling of texture distributions, leading to more realistic results but failing to capture large-scale structure, especially for non-stationary textures. Generative adversarial networks (GANs) are also widely adopted in texture synthesis [51, 52, 68]; for instance, Zhou *et al.* [68] used a self-supervised approach to expand a single texture by learning the internal patch distribution, achieving impressive detail and structure expansion. Recently, Self-Rectification [70] leverages pre-trained diffusion networks and the self-attention mechanism to gradually refine a lazy-

editing input, addressing the intricate challenge of synthesizing non-stationary textures.

Neural style transfer applies the artistic style of a source image to a new content image. Gatys *et al.* [22] pioneered this field by using Gram loss as style representation and proposed a neural style transfer algorithm that combines content and style loss in the optimization. While Gram loss and its variants [27, 30, 39] are the most widely used style losses, they primarily assess global distribution differences and cannot account for local semantic correspondences. Approaches like CNNMRF [40] and Contextual Loss [43] address this by computing semantic similarity in high-dimensional feature spaces to enable semantically coherent style transfer. Later works incorporated transformers with self-attention to capture stronger relationships between style and content. Deng *et al.* [12] introduced a fully transformer-based architecture, StyTR², achieving state-of-the-art results at the time. Recent advances in diffusion models enable interpretable and controllable content-style separation. InST [66] proposed inversion-based learning of artistic style from a single painting. StyleDiffusion [60] introduced a CLIP-based style disentanglement loss. StyleID [10], a training-free method, manipulates the self-attention features of a pre-trained diffusion model by substituting the keys and values of the content with those of the style image in cross-attention mechanisms.

Appearance transfer, a specialized semantic style transfer, aims at transferring the appearance of semantically corresponding regions. Early works [28, 45, 71] use paired or unpaired datasets to train GANs for domain-specific appearance transfer. Tumanyan *et al.* [57] extract structure and appearance features using a pre-trained DINO-ViT [8], and trains a generator for each image pair. Recently, cross-image attention [2] with pre-trained diffusion models realizes zero-shot appearance transfer by implicitly establishing semantic correspondences across images.

Customized/Personalized image generation based on text-to-image (T2I) diffusion models has attracted particular attention recently, which aims to learn the style from one or more reference images to generate new content. Fine-tuning approaches [1, 18–20, 31, 36, 50, 54] enable models to learn novel style concepts from a few images. However, these methods are prone to overfitting, potentially resulting in degraded image quality or content leakage. To alleviate this issue, B-LoRA [19] leverages LoRA [26] (Low-Rank Adaptation) to implicitly separate the style and content components of a single image. Pair Customization [31] learns stylistic differences from a single image pair and then applies the acquired style to the generation process. Encoder-based methods [9, 33, 59, 61, 63, 64] utilize visual encoders to capture image information and establish mappings between image prompts and models through huge dataset training. While currently favored as state-of-the-

art, these techniques are constrained by the capabilities of visual encoders, often extracting only abstract style information and struggling with fine-grained textures of the reference. Several works have proposed plug-and-play solutions for training-free style customization. For example, StyleAligned [25] and Visual Style Prompt [29] maintain style consistency by preserving the queries from the original features while sharing or swapping the keys and values with those from reference features in the late self-attention layers. RB-Modulation [49] modulates the drift field of reverse diffusion dynamics by incorporating desired attributes (*e.g.*, style or content) through a terminal cost.

3. Method

3.1. Preliminaries

Latent diffusion models (LDM), exemplified by Stable Diffusion [46, 48], have achieved state-of-the-art performance in image generation due to the robust ability to model complex data distributions. In LDM, an image x is first compressed into a learned latent space using a pretrained VAE $\mathcal{E}(\cdot)$. A UNet-based denoising network $\epsilon_\theta(\cdot)$ is subsequently trained to predict the noise during the diffusion process by minimizing the mean squared error between the predicted noise and the actually added noise ϵ :

$$\mathcal{L}_{\text{LDM}} = \mathbb{E}_{z \sim \mathcal{E}(x), y, \epsilon \sim \mathcal{N}(0,1), t} \left[\|\epsilon_\theta(z_t, t, y) - \epsilon\|_2^2 \right], \quad (1)$$

where y denotes the condition and t represents the timestep. The denoising UNet typically consists of a bunch of convolution blocks and self-/cross-attention modules, all integrated within the predictive branch of *residual* architecture.

KV-injection is widely employed in image editing [2, 7, 58], style transfer [10, 25, 29], and texture synthesis [70]. It is built upon the self-attention mechanism and uses the self-attention features in diffusion models as plug-and-play attributes. The self-attention mechanism is formulated as:

$$\text{Self-Attn}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d}}\right)V. \quad (2)$$

At the core of the attention mechanism lies the calculation of a weight matrix based on the similarity between the queries (Q) and keys (K), which is used to perform a weighted aggregation of the values (V). KV-injection extends this mechanism by copying or sharing the KV features across different synthesis branches. Its key assumption is that KV features represent the visual appearance of an image. During sampling, replacing the KV features in the synthesized branch with those KV from the corresponding timestep of the exemplar can realize appearance transfer from the source image to the synthesized target.

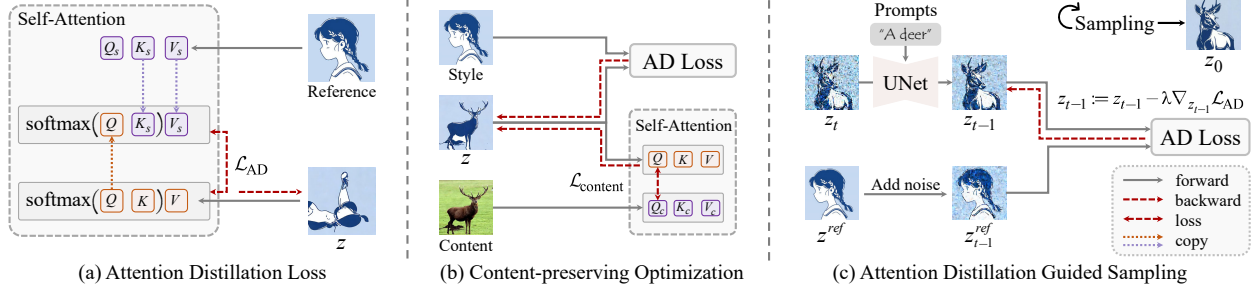


Figure 2. **Overview of attention distillation.** Based on the self-attention mechanism in diffusion models, we compute the difference between the ideal and the current stylization, formulating a novel Attention Distillation (AD) loss (a). The new loss acts like a style loss. When combined with a content loss (also derived from the self-attention mechanism), we can realize high-quality content-preserving synthesis, such as style transfer or appearance transfer (b). Our attention distillation loss can be incorporated into the normal diffusion sampling process as an improved Classifier Guidance (c), which enables a broad scope of example-based image generation applications.

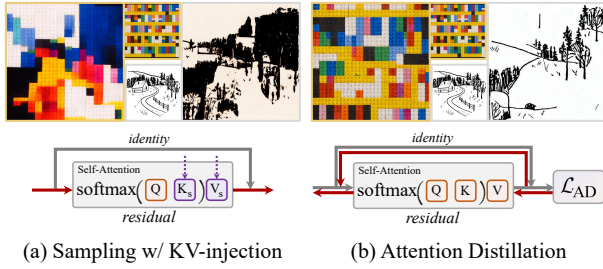


Figure 3. **Differences between KV-injection and attention distillation.** We start with the same latent for sampling and optimization, both running 100 steps, using empty prompts. The information flow (red arrows) differs only from the identity connection. However, the results of our attention distillation optimization (b) are clearly superior to sampling with KV-injection (a).

3.2. Attention Distillation Loss

Although KV-injection has achieved noticeable results, it falls short in preserving the style or texture details of the reference due to the residual mechanism; see Fig. 3 (a) for example. KV-injection only operates on the residual, which means the information flow (red arrows) is subsequently influenced by the identity connection, leading to an incomplete transfer of information. As a result, the sampling outputs cannot fully reproduce the desired visual details.

In this work, we propose a novel loss function to distill visual elements by reaggregating features within the self-attention mechanism; therefore, we refer to it as Attention Distillation (AD) loss. We leverage the UNet of the pre-trained T2I diffusion model, Stable Diffusion [48], to extract image features from self-attention modules. As illustrated in Fig. 2 (a), we first reaggregate the visual information of the KV features (K_s and V_s) from the reference branch according to Q from the target branch, which is the same as KV-injection. We regard this attention output as the *ideal* stylization. Then, we calculate the attention output of the target branch and compute the L1 loss w.r.t. the ideal

attention output, which defines the AD loss:

$$\mathcal{L}_{AD} = \|\text{Self-Attn}(Q, K, V) - \text{Self-Attn}(Q, K_s, V_s)\|_1. \quad (3)$$

We can use the proposed AD loss to optimize a random latent noise via gradient descent, resulting in vivid texture or style reproduction in the output; see Fig. 3 (b) for example. This can be attributed to the backpropagation in optimization, which allows information to flow not only across the (*residual*) self-attention modules but also through the identity connection. With continuous optimization, the gap between Q and K_s gradually narrows, making attention more and more accurate, and eventually, features are correctly aggregated to produce the desired visual details.

Following recent experimental analysis [7, 29, 58], we empirically select the last 6 self-attention layers of the UNet to compute AD loss. Additionally, during optimization, we simulate the sampling process of diffusion models by linearly decreasing the timestep t input to the UNet from T to 0. We begin with different random latent noises and optimize them over 100 steps. Note that during the whole optimization, the predicted noise from U-Net is totally discarded, and we continuously update the same latent.

To better understand our AD loss, we present the optimization results across multiple runs, as shown in Fig. 4. These results demonstrate that: i) AD loss effectively distills high-quality visual characteristics in style and texture; ii) AD loss is self-adapted to different spatial structures, showcasing diversity across multiple runs.

3.3. Content-preserving Optimization

With the texture and style distilled by AD loss, we can further align the synthesized content to another reference image using a content loss. Such optimization allows the synthesis of images that transform the visual elements of one image while preserving the target content, achieving tasks such as style transfer, appearance transfer, and more.

As illustrated in Fig. 2 (b), we define the content loss

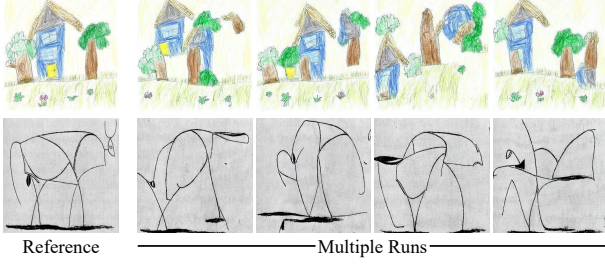


Figure 4. **Optimizing attention distillation loss across multiple runs.** The coherence in texture and style, and the variations in structure across multiple runs of the same reference, demonstrates the ability of our AD loss in style alignment and spatial adaption.

similarly to AD loss, which is also based on the self-attention mechanism, fully drawing the advantage of the deep understanding of images in diffusion models. In particular, the L1 loss computed between the target queries Q and the reference queries Q_c , formulates the content loss:

$$\mathcal{L}_{\text{content}} = \|Q - Q_c\|_1. \quad (4)$$

In implementation, we also select the last 6 self-attention layers to compute the content loss, consistent with AD loss. The objective of content-preserving optimization is:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{AD}} + \lambda \mathcal{L}_{\text{content}}. \quad (5)$$

After optimization, the optimized latent code is decoded into image space using the pretrained VAE. We have summarized the content-preserving optimization in Algorithm 1 of the supplementary material.

3.4. Attention Distillation Guided Sampling

The syntheses above are using backpropagation optimization. In this section, we introduce how we incorporate attention distillation loss into the sampling process of diffusion models in an improved Classifier Guidance manner.

According to [13], Classifier Guidance alters the denoising direction during the denoising process, thereby generating samples from $p(z_t|c)$, which can be formulated as:

$$\hat{\epsilon}_\theta = \epsilon_\theta(z_t, t, y) - \alpha \sigma_t \nabla_{z_t} \log p(c|z_t), \quad (6)$$

where t is the timestep, y denotes the prompts, and ϵ_θ and z_t refer to the denoising network and the latent in LDM, respectively. α controls the guidance strength. Inspired by [17], we guide the diffusion sampling process using an energy function based on attention distillation loss.

Specifically, during DDIM sampling [55], the latent z_t at timestep t are translated to z_{t-1} according to the update direction ϵ_θ estimated from the denoising network:

$$z_{t-1} = \sqrt{\bar{\alpha}_{t-1}} \hat{z}_0 + \sqrt{1 - \bar{\alpha}_{t-1}} \epsilon_\theta, \quad (7)$$

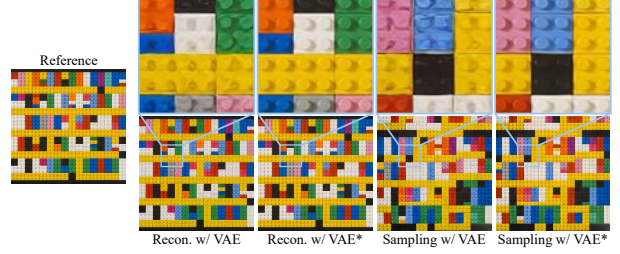


Figure 5. **Improved VAE decoding.** The pretrained VAE is lossy in high-frequency details. Fine-tuning the VAE with the reference image over several steps (denoted as VAE*) can enhance the reconstruction quality and the decoding for novel image synthesis.

where $\hat{z}_0 = \frac{z_t - \sqrt{1 - \bar{\alpha}_t} \epsilon_\theta}{\sqrt{\bar{\alpha}_t}}$. Replace ϵ_θ with $\hat{\epsilon}_\theta$ from Eq. (6):

$$z_{t-1} = \underbrace{\sqrt{\bar{\alpha}_{t-1}} \hat{z}_0 + \sqrt{1 - \bar{\alpha}_{t-1}} \epsilon_\theta}_{\text{DDIM Sampling}} + \alpha C \cdot \nabla_{z_t} \log p(c|z_t), \quad (8)$$

where C is a constant related to t . We define the energy function using our AD loss by substituting $\nabla_{z_t} \log p(c|z_t)$ with $\nabla_{z_t} \mathcal{L}_{\text{AD}}$. However, we found configuring the guidance strength is tricky; see Sec. 4 for detailed experimental analysis. To address this issue, we introduce an Adam optimizer [35] to automatically manage the strength and compute gradients. For simplification, we approximate the term $\nabla_{z_t} \mathcal{L}_{\text{AD}}$ by $\nabla_{z_{t-1}} \mathcal{L}_{\text{AD}}$, enabling an initial DDIM sampling for z_{t-1} , followed by a straightforward optimization update. The AD loss takes as inputs the latent z_{t-1} and the noise-disturbed reference latent z_{t-1}^{ref} , as illustrated in Fig. 2 (c). We update z_{t-1} via AD loss optimization as:

$$z_{t-1} := z_{t-1} - \alpha C \cdot \nabla_{z_{t-1}} \mathcal{L}_{\text{AD}}(z_{t-1}, z_{t-1}^{\text{ref}}). \quad (9)$$

With the guidance from AD loss, we can compute the losses on the latents with timestep conditioning, rather than converting the latent to image space and calculating image-level losses, as done in recent works [4, 41, 49]. Using the Adam optimizer also enables us to establish a universal learning rate, alleviating the challenge of setting the guidance strength. The attention distillation guided sampling is detailed in Algorithm 2 of the supplementary. Note that the content loss proposed in Eq. (4) can also be added into the sampling process working with AD loss, further to preserve the structure of a content reference image.

3.5. Improved VAE Decoding

Experimental evidence [3, 72] suggests that the VAE employed in latent diffusion models is perceptually lossy. For tasks that require high-frequency local details, such as texture synthesis, we can *optionally* fine-tune the weights θ of the VAE decoder $\mathcal{D}(\cdot)$ on the example image x to enhance its reconstruction quality using L1 loss, following [3]:

$$\theta^* = \arg \min_{\theta} \|\mathcal{D}_\theta(\mathcal{E}(x)) - x\|_1, \quad (10)$$

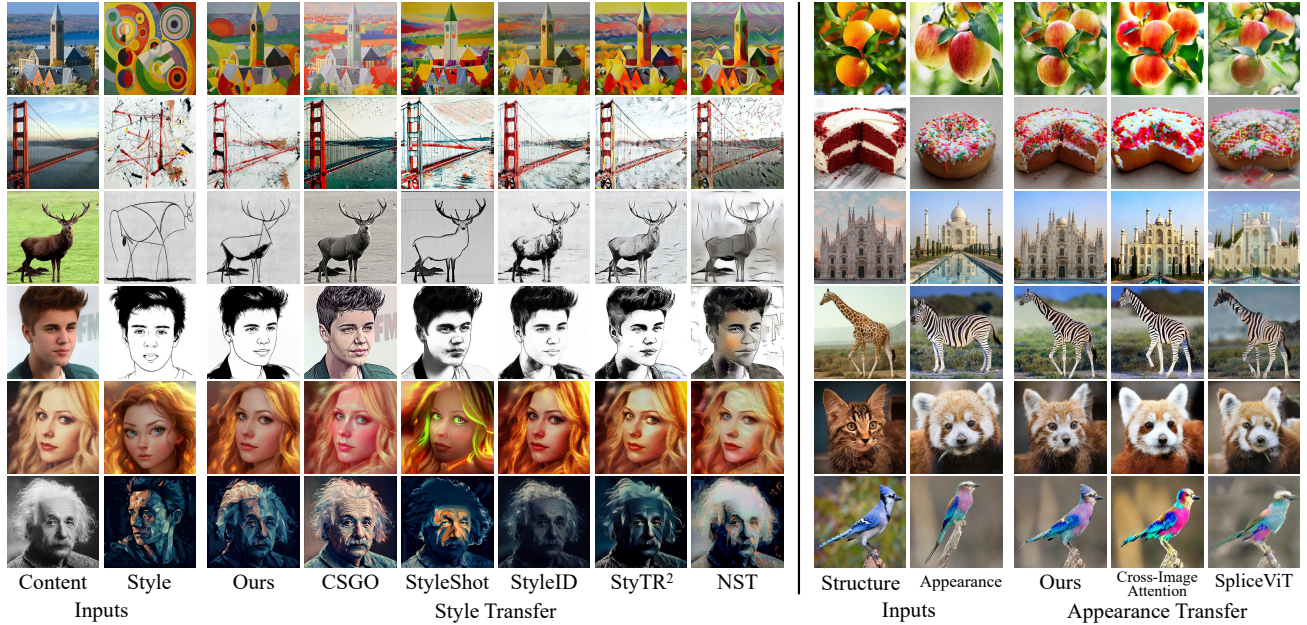


Figure 6. **Comparisons of style and appearance transfer.** Our comparisons primarily focus on recent diffusion-based methods of style and appearance transfer, including CSGO [63], StyleShot [33], StyleID [10], and Cross-Image Attention [2]. Additionally, we include traditional methods such as NST [22] and transformer-based methods such as StyTR2 [12] and SpliceViT [57] for comparison.

where $\mathcal{E}(\cdot)$ denotes the VAE encoder. Fig. 5 presents some results of the reconstruction and sampling, showing improved perceptual quality with the fine-tuned VAE.

4. Experiments

4.1. Applications and Comparisons

In the following, we apply our attention distillation loss to various visual characteristic transfer tasks and compare the results to state-of-the-art methods in each application. See the supplementary materials for detailed parameter configurations, running time, and additional results.

Style and Appearance Transfer. Following the spirit of the prominent work of Gatys *et al.* [22], we achieve style and appearance transfer through the optimization method described in Sec. 3.3. We compare our method to CSGO [63], StyleShot [33], StyleID [10], StyTR² [12] and NST [22] for style transfer, as well as Cross-Image Attention [2] and SpliceViT [57] for appearance transfer. Fig. 6 presents the qualitative comparison results. In style transfer, our method effectively captures high-quality, coherent style characteristics while simultaneously preserving the semantic structures of the content image. This is particularly evident in the sketch styles in the 3rd and 4th rows. In contrast, baseline methods exhibit notable style discrepancies, despite retaining the original structure. In appearance transfer, our method also shows superiority, avoiding the over-saturation of color seen in Cross-image Attention.

Style-specific Text-to-Image Generation. As described in Sec. 3.4, we can apply our AD loss within the diffusion sampling, thereby realizing style-specific text-to-image generation. We set the reference as the desired style image. Fig. 7 showcases some generated results, along with a comparison to alternative methods, including Visual Style Prompting [29], InstantStyle [59], and RB-Modulation [49]. The results in Fig. 7 demonstrate that our method aligns textual semantics comparably to existing methods while achieving notably better style coherence with the reference.

In addition to the above approach, we further incorporate ControlNet [65] to enable style-specific text-to-image generation with additional conditioning on various modalities, *e.g.* depth and canny edges. Fig. 8 presents some generated examples. More results can be seen in the supplementary.

Controlled Texture Synthesis. Our method can be applied to texture optimization as demonstrated in Sec. 3.2. Inspired by [7], we further incorporate mask guidance when calculating attention distillation loss, thereby constraining the values queried by Q , resulting in controlled texture synthesis. Specifically, given a source texture $x_s \in \mathbb{R}^{h \times w \times 3}$, its corresponding source segmentation map $S^s \in \mathbb{R}^{h \times w \times 1}$, and the target segmentation map $S^t \in \mathbb{R}^{h' \times w' \times 1}$, we first flatten S^s and S^t and downsample them to match the resolution of attention features, resulting in $\tilde{S}^s$ and $\tilde{S}^t$. Then, we compute the guiding mask M ; with this mask, we restrict the visual information aggregated by Q in attention computation to focus solely on the corresponding regions:

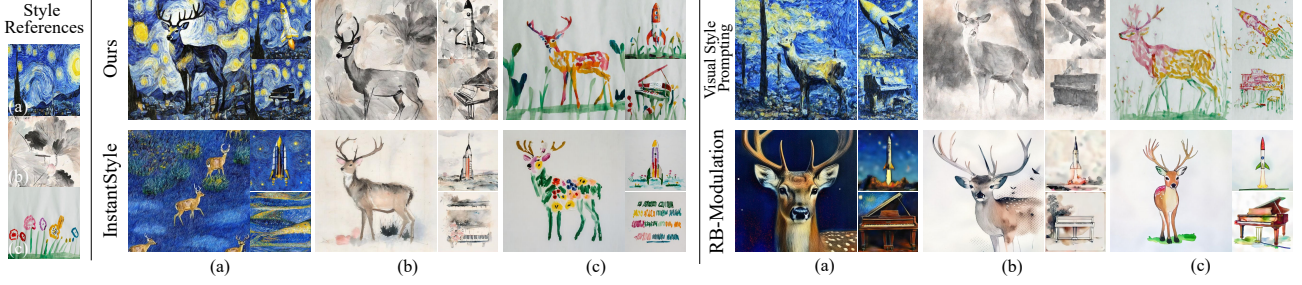


Figure 7. **Comparisons of style-specific text-to-image generation.** We compare our approach to InstantStyle [59], Visual Style Prompting [29], and RB-Modulation [49] using three style references as examples. For each style, we utilize the same text prompts: “A deer” (left), “A rocket” (top right), and “A piano” (bottom right).



Figure 8. By combining attention distillation guided sampling with ControlNet [65] into the text-to-image pipeline, we can produce high-quality results that align the structure with the condition image while maintaining style coherence with the style reference.

$$M_{i,j} := \begin{cases} \text{True} & \text{if } \tilde{S}_i^s = \tilde{S}_j^t \\ \text{False} & \text{otherwise} \end{cases} \quad (11)$$

$$\mathcal{L}_{\text{MAD}} = \|\text{Self-Attn}(Q, K, V) - \text{Self-Attn}(Q, K_s, V_s; M)\|_1 \quad (12)$$

To further ensure the alignment with the target mask, we fill the target mask as initialization with pixels randomly drawn from corresponding labeled regions of the source image. Then, we regard this init as the content reference and add a content loss as Sec. 3.3 based on the query features. Figs. 1 and 9 present our controlled texture synthesis results. Compared to GCD [69], a patch-based neural texture optimization method, our results exhibit comparable texture details with smoother object edges. In contrast, GCD suffers from artifacts of color aliasing; see the 2nd row of Fig. 9.

Recently, Self-Rectification [70] introduced a “lazy-editing” control for generating non-stationary textures. Aiming at the same goal, we utilize SDEdit [44] to preserve the structure of the layout image edited by the user. Then, we incorporate our proposed AD loss and content loss into the sampling (as Sec. 3.4). As compared in Fig. 9, Self-Rectification outputs smoother texture transitions, while our results better adhere to the original texture examples.

Texture Expansion. It is very difficult to synthesize ultra-high resolution textures using traditional methods, given the limited patch sources. Here, we apply our attention distillation guided sampling to the MultiDiffusion [5] model, enabling texture expansion to arbitrary resolution. Although

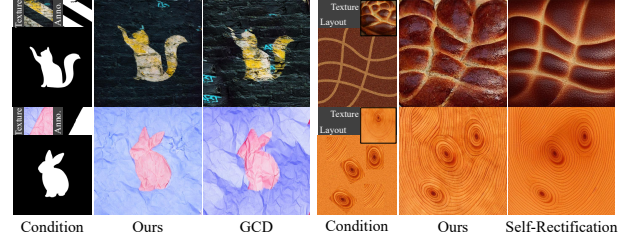


Figure 9. **Comparisons of controlled texture synthesis.** Left: annotation control; Right: layout control.

SD-1.5 [48] is trained on images of size 512×512 , surprisingly, it demonstrates robust capabilities in large-size texture synthesis when incorporating attention distillation. Fig. 10 presents comparisons of texture expansion to size 512×1536 with GCD [69] and GPDM [16]. Our approach shows significant advantages in such a challenging task.

4.2. Ablation Studies

In this section, we present ablation study results on two aspects of our approach: i) the impact of content loss weight in content-preserving optimization (Sec. 3.3), and ii) the optimizer for managing guidance strength in attention distillation guided sampling (Sec. 3.4).

Content loss weight. As shown in Fig. 11, varying the content loss weight λ brings intriguing effects on the transfer results. In style transfer, for instance, the abstract style example shown on the left illustrates how the adjustment of λ results in different levels of abstraction, offering flexibility for artistic creation. In appearance transfer, thanks to the precise semantic understanding from diffusion networks, the facial image transfer shown on the right exhibits a smooth identity transition along with the λ varies.

Optimizer. Fig. 12 demonstrates the importance of the optimizer in managing guidance strength. We experimentally test a naive strategy of manually setting the guidance strength to control the scale of gradient updates to the latents. However, varying this strength manually often fails to yield reasonable results: textures or appearance features in the examples are typically lost, as shown in the last three

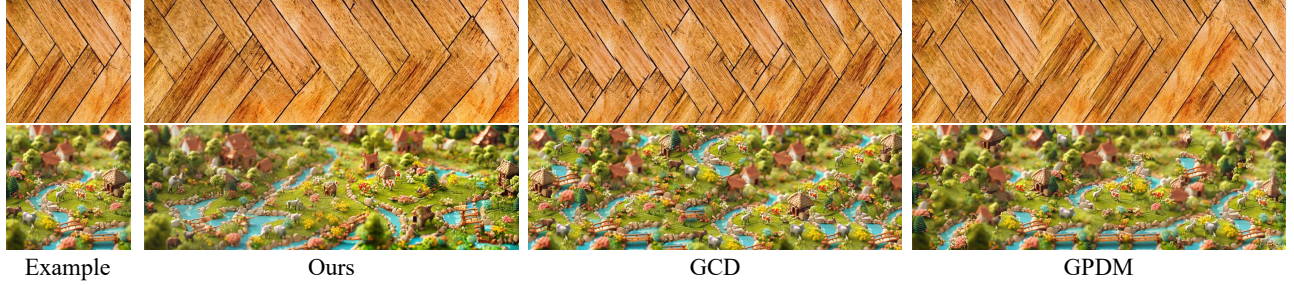


Figure 10. **Comparisons of texture expansion.** Taking a 512×512 texture image as the input example, we expand it with automatic semantic understanding and synthesize results at the resolution 512×1536 . We compare our approach with GCD [69] and GPDM [16].

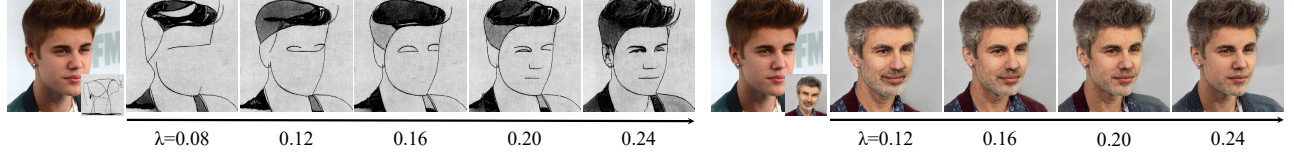


Figure 11. **Ablation of content loss weights.** Varying the content loss weights λ on two examples, one each for style transfer (left) and appearance transfer (right). The results demonstrate how the content loss helps preserve the source image content during the transfer. When the weights are within an appropriate range, the results exhibit varying levels of abstraction and appearance transition.

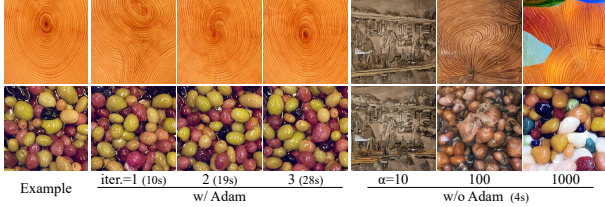


Figure 12. **Impact of the optimizer for managing guidance strength.** Both cases—with and without the Adam optimizer [35]—were conducted with 50 diffusion sampling steps. All results are generated using Stable Diffusion v1.5 [48], with empty text prompts. The learning rate of the Adam optimizer is set to 0.02.

columns of Fig. 12. In contrast, introducing an Adam optimizer to manage latent optimization produces results that closely match the visual characteristics of the input examples (columns 2~4 in Fig. 12). Furthermore, increasing the number of optimization iterations within each timestep during sampling generally enhances the quality of generated results, although it also brings extra computation time. In practice, we set the iteration number to 2 to achieve an efficient balance, delivering high-quality results effectively.

4.3. User Preference Study

To validate the qualitative analysis, we conduct a user study with 30 questions (5 questions for each of 6 selected competitors) on three transfer tasks. In each question, we show-case two results: one from our method and one from a competitor. The user was asked to choose the better one based on the provided instructions and criteria. We have collected 1500 responses from 50 participants, and the overall preference score is summarized in Fig. 13. Our method consistently outperforms the alternatives by significant margins. Please refer to the supplementary for more details.

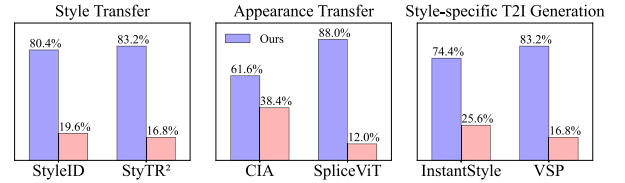


Figure 13. **User preference score.** We report the overall preference score comparing our method to selected alternatives across three transfer tasks. Each comparison is conducted individually, directly evaluating user preference between our results and those of each competing method. Note CIA stands for Cross-Image Attention [2] and VSP as [29]. See supplementary for more details.

5. Conclusion

We have presented a unified approach for various visual characteristics transfer tasks, including style/appearance transfer, style-specific image generation, and texture synthesis. The key to the proposed method is a novel attention distillation loss, which calculates the difference between the ideal and current stylization, and gradually modifies the synthesis. Our method overcomes the limitations of previous works, and experiments have validated its superiority.

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